MNS Department
Semester: Summer-2025; Course ID: PHY 111
Course Title: Principles of Physics I; Total Marks: 15

Assignment-2

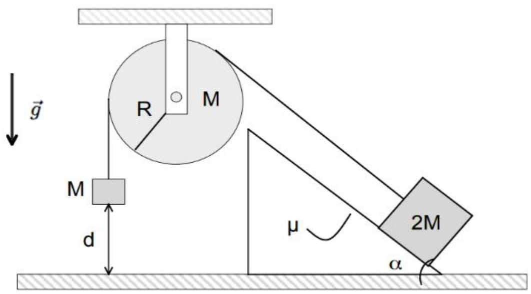
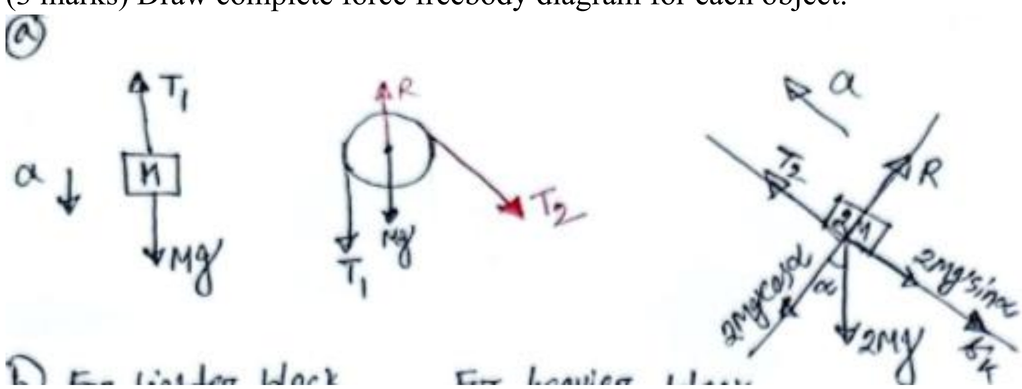


Fig. 1

1. An Atwood machine shown in fig. 1 consists of a fixed pulley wheel of radius $R = 50\ cm$ and uniform mass $M = 1.0\ kg$ (a disk), around which an effectively massless string passes connecting two blocks of mass M and $2M$. The lighter block is initially positioned a distance, d above the ground. The heavier block sits on an inclined plane with opening angle $\alpha = 15°$. There is a coefficient of friction $\mu = 0.15$ between the surfaces of this block and the inclined plane. Constant gravitational force acts downwards, assume that the string never slips, and lighter block moves down.

   a) (3 marks) Draw complete force freebody diagram for each object.



b) (9 marks) Determine the angular acceleration of the disk. (show detail calculation)

(b) For lighter block

$T_1 - Mg = -Ma$

$\Rightarrow T_1 = 1\times 9.8 - 1a$

$\Rightarrow T_1 = 9.8 - a$ —— (I)

For heavier block

$x \rightarrow T_2 - 2Mg\sin\alpha - f_k = 2Ma$

$\Rightarrow T_2 - 2Mg\sin\alpha - \mu R = 2Ma$

$\Rightarrow T_2 - 2\times 1\times 9.8\sin 15^\circ - 0.15\times 18.93 = 2\times 1a$

$\Rightarrow T_2 = 7.91 + 2a$ —— (II)

$y \rightarrow R - 2Mg\cos\alpha = 0$

$\Rightarrow R = 2\times 1\times 9.8\times\cos 15^\circ$

$\therefore R = 18.93\,N$

For the disk

$\vec{\tau}_{net} = I\vec{\alpha}$

$\Rightarrow \vec{\tau}_1 + \vec{\tau}_2 = I\vec{\alpha}$

$\Rightarrow 0.5T_1 - 0.5T_2 = \frac{1}{2}MR^2\left(\frac{a}{R}\right)$

$\Rightarrow 0.5(9.8 - a - 7.91 - 2a) = \frac{1}{2}\times 1\times(0.5)^2\frac{a}{0.5}$

$\Rightarrow 1.89 - 3a = \frac{a}{2}$

$\Rightarrow a = 1.89\times\frac{2}{7} = \boxed{0.54\,ms^{-2}}$

$\therefore \alpha = \frac{a}{R} = \frac{0.54}{0.5} = \boxed{1.08\,rad\,s^{-2}}$

$\tau_1 = RT_1\sin 90^\circ$
$= 0.5\times T_1\times 1$
$= 0.5T_1$ (outward)

$\tau_2 = RT_2\sin 90^\circ$
$= 0.5T_2\times 1$
$= 0.5T_2$ (Inward)

c) (3 marks) Find the tensions in the string and amount of net torque acting on the disk.

(c) From eqn (I) $T_1 = 9.8 - 0.54 = 9.26\,N$

" " (II) $T_2 = 7.91 + (2\times 0.54) = 8.99\,N$

and net torque, $\tau_{net} = 0.5(9.26 - 8.99) = 0.135\,Nm$ (outward)

or

$\tau_{net} = I\alpha$
$= \frac{1}{2}MR^2\alpha$
$= \frac{1}{2}\times 1\times(0.5)^2\times 1.08$
$= 0.135\,Nm$ (outward)